Prof. Stefan Minner
Principal Editor, *International Journal of Production Economics*

Submission of an original research article
Replenishing perishables on shared delivery routes: Stock-age-based inventory policies, stochastic lookahead and decision-focused learning

Dear Professor Minner,

I am pleased to submit the above manuscript for consideration as a research article in the *International Journal of Production Economics*.

Fresh food is replenished daily, its stock ages every night, and whatever is not sold within its shelf life is thrown away. In retail distribution, deliveries to different stores also share a small fleet, so a store that is not on today's route cannot be replenished today. The manuscript asks how contextual demand forecasts should be turned into replenishment decisions in this setting. It studies a periodic-review perishable inventory system with FIFO issuing, lost sales and shelf capacity, in which each day's replenishment is shipped on shared, capacitated routes, and compares in closed loop the inventory rules recommended in the perishable inventory literature, the deliver-up-to-level rule of Crama et al. (2018), and deterministic, robust and scenario-based lookahead policies, including the Scenario Predict-then-Optimize method of Jia et al. (2025). All policies observe the same age-structured stock and use the same forecasts.

Contributions.

- *Decision-focused learning of perishable order-up-to levels.* The proposed policy, UL-DFL, keeps the visit trigger of the deliver-up-to-level rule and learns context- and stock-age-dependent order-up-to levels from the realised holding, waste and lost-sales cost of whole episodes. On fresh test episodes of the base case it is 1.1% cheaper than the rule tuned on a fine grid (Holm-adjusted $p < 0.001$) and wastes 28% less at nearly the same fill rate, which no parameterisation of the rule achieves. Across thirteen settings it is significantly cheaper than the rule in nine and wastes less in twelve. The learned corrections read as inventory rules: lower order-up-to levels on promotion days and after the weekly demand peak.

- *An inventory-centred benchmark.* Twelve policies are evaluated in one closed-loop simulator, with every lookahead decision obtained from a MILP with exact route costs and solved to a proven gap. Outcomes are reported in inventory terms (cost composition, fill rate, waste rate, delivery frequency and store-level service) across network layouts, shelf life, the lost-sales/waste cost ratio, fleet capacity, routing cost, demand variability, issuing policy and observed fresh-food demand from a public retail dataset (Wang et al., 2025).

- *Structural results.* The inventory recourse admits FIFO-optimal solutions once early disposal is allowed, the marginal value of a delivery is bounded by the lost-sales and waste penalties, and the stationary points of decision-focused training reduce to the perishable newsvendor fractile. All results are verified numerically.

- *Managerial insights.* Well-tuned stock-age-based rules with optimally routed orders are 2–12% cheaper than integrated inventory-routing policies unless routing is expensive; deterministic plans over-stock, and a day-specific safety quantile repairs this; and integrated plans concentrate stock-outs at the stores that are most expensive to visit.

Fit with the journal. The study takes the perishable inventory management perspective developed in this journal, from inventory control of perishables in supermarkets (van Donselaar et al., 2006) to the value of stock-age information in ordering (Haijema, 2014; Haijema and Minner, 2019), and extends it to stores whose deliveries share vehicles, a setting so far studied

mainly from the routing side (Soysal et al., 2015). It also connects this perspective with the growing literature on learning-based inventory decisions (Kaynov et al., 2024).

Rigour and transparency. All parameters are tuned on separate validation episodes, the primary hypotheses are prespecified and tested on fresh test episodes that played no role in the design, tuning or training of any policy, and p-values are Holm-adjusted. Negative results are reported as such: on observed fresh-food demand the learned corrections did not transfer to new products, and decision-focused learning on top of the deterministic lookahead is matched by a simpler policy with day-specific quantile levels. The preparation of the real data is described in full, and an anonymised replication package with the code, seeds and result files reproduces every table and figure.

Declarations. The manuscript is original, solely authored, and has not been published and is not under consideration elsewhere. I have no competing interests, and the work received no specific funding. The study involves no human participants or animal subjects. The use of generative AI (Claude, Anthropic) is disclosed in the manuscript and in the declarations file, which also contains the CRediT statement and the data and code availability statement. The manuscript file is anonymised; author details are given on the separate title page.

Thank you for considering this manuscript. I look forward to hearing from you.

Sincerely,

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